Introduction to metabolism

Metabolism is the sum of all the chemical reactions that take place in cells. It is a coordinated and regulated process that performs the following functions:

- To obtain and store chemical energy.
- To convert nutrients into the body's own molecules and precursors for macromolecule synthesis.
- To synthesize macromolecules, including proteins, nucleic acids, lipids, and polysaccharides, as well as other cellular components.

Forms of obtaining carbon and energy

Living organisms can be divided into two large groups based on the chemical form from which they obtain carbon for the synthesis of their molecules:

- **Autotrophs** use inorganic carbon, mainly **carbon dioxide** (CO_2), to synthesize their organic molecules. Plants and many photosynthetic microorganisms are autotrophic organisms.
- **Heterotrophs** obtain carbon from **organic compounds** produced by other organisms. Animals, fungi, and many microorganisms are heterotrophic organisms.

Therefore, heterotrophs ultimately depend on organic compounds produced by autotrophic organisms.

Organisms can also be classified according to their source of energy. **Phototrophs** obtain energy from **light**, whereas **chemotrophs** obtain energy from the **oxidation of chemical compounds**.

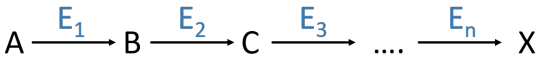
These two classifications can be combined. **Photoautotrophs**, such as plants, use light as their energy source and use CO_2 as their main carbon source. **Chemoautotrophs** obtain energy by oxidizing inorganic compounds and use CO_2 as their carbon source. **Chemoheterotrophs** obtain both carbon and energy from organic compounds.

Humans are chemoheterotrophs: we obtain both carbon and energy from organic nutrients.

Metabolic pathways

The reactions that make up metabolism are organized into metabolic pathways.

A **metabolic pathway** is a sequence of enzyme-catalyzed reactions that transforms one or more initial substrates into a final product through a series of small chemical changes. The intermediate compounds formed along the pathway are called **metabolites**.



Although there are thousands of different reactions in metabolism, there are only a few types of chemical reactions that repeat throughout the entire metabolism:

- Transfer of functional groups
- Oxidation–reduction reactions
- Isomerization reactions
- Hydrolysis and condensation reactions
- Formation and cleavage of carbon–carbon bonds

Central metabolic pathways are highly conserved among different living beings. These pathways participate in the degradation and synthesis of key metabolites and in the capture, transfer, and storage of metabolic energy. **Central metabolism** forms the core of intermediary metabolism, which comprises the pathways involving low-molecular-mass precursors, intermediates, and products. The processes of DNA replication, transcription, and protein synthesis are not usually considered part of intermediary metabolism. The pathways of central metabolism are:

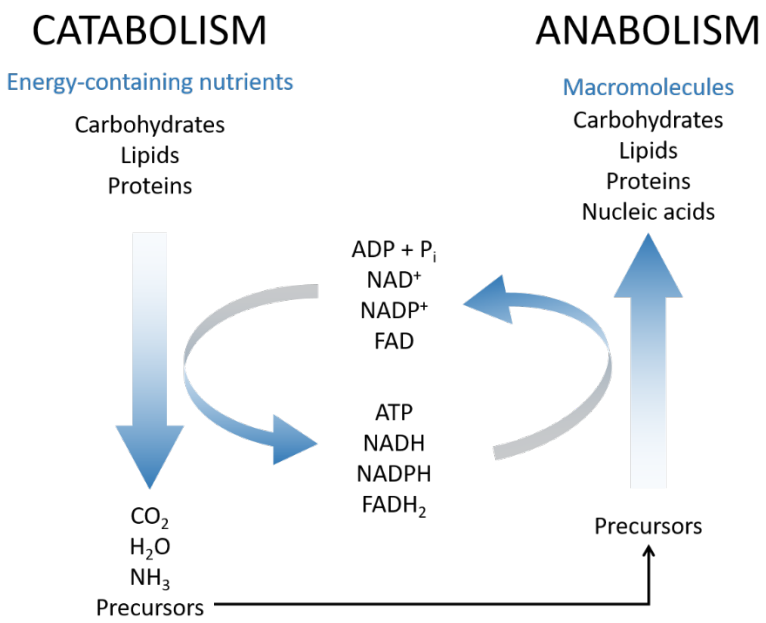
- Carbohydrate metabolism:
  - Glycolysis
  - Pentose phosphate pathway
  - Gluconeogenesis
  - Glycogen breakdown
  - Glycogen synthesis
- Lipid metabolism:
  - Fatty acid degradation
  - Fatty acid and lipid synthesis
- Amino acid metabolism:
  - Amino acid degradation
  - Amino acid synthesis
- Nucleotide metabolism:
  - Degradation of nucleotides
  - Synthesis of nucleotides
- Final oxidation and energy production:
  - Citric acid cycle
  - Oxidative phosphorylation

Phases of metabolism

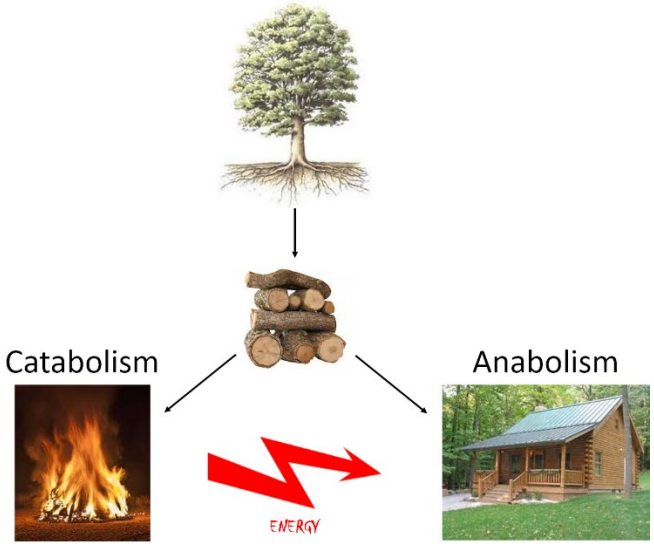
Metabolism is subdivided into two phases:

- **Catabolism** is the degradative phase of metabolism. Nutrients obtained from the environment or stored in the body are broken down into smaller molecules and metabolic intermediates and, ultimately, into products such as CO_2, H_2O, and NH_3. The energy released during these reactions can be captured in ATP and in reduced electron carriers such as NADH and $FADH_2$.

- **Anabolism** is the biosynthetic phase of metabolism. Small precursors are combined and modified to form complex molecules such as carbohydrates, lipids, proteins, and nucleic acids. Anabolic pathways require energy, usually supplied by ATP, and reducing power, commonly supplied by NADPH.



The relationship between catabolism and anabolism can be illustrated using wood from a tree. Once the wood has been obtained, it can be burned to release energy or used as a material to build a cabin. Similarly, nutrients can be degraded to release energy or used as sources of carbon for the synthesis of cellular molecules. Biosynthesis itself also requires part of the energy obtained through catabolism.



The analogy can be extended because both **combustion** and **aerobic catabolism** involve the oxidation of carbon compounds to carbon dioxide:

$$Glucose\ (C_6H_{12}O_6) + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + energy$$

However, an important difference is that cells oxidize nutrients through a controlled sequence of enzyme-catalyzed reactions, allowing part of the released energy to be captured in ATP.

In general, **catabolic pathways are convergent**: many different nutrients are degraded to form a relatively small number of common metabolites. In contrast, **anabolic pathways are divergent**: a limited number of precursors are used to synthesize a wide variety of cellular molecules. Some metabolic pathways have both catabolic and anabolic functions and are therefore called **amphibolic pathways**.

Stages of catabolism

In the case of catabolism, three stages are usually distinguished:

1. In the **first stage**, macromolecules are broken down into their **constituent units**.

   Proteins ⇒ Amino acids

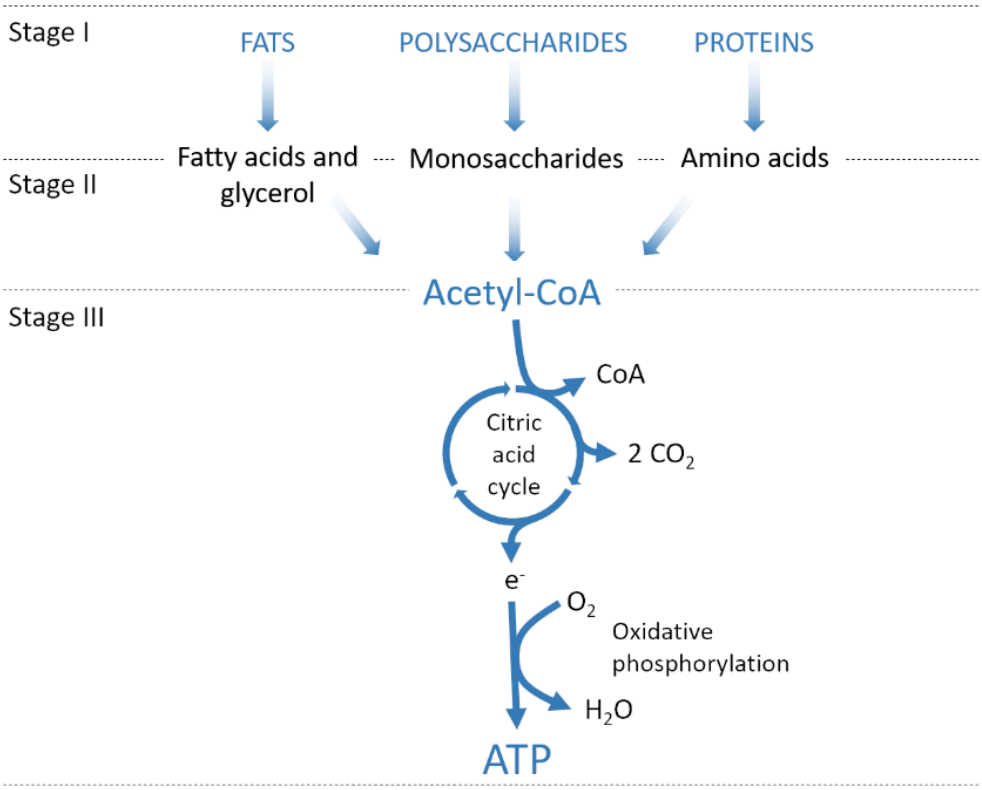
   Polysaccharides ⇒ Monosaccharides

   Triacylglycerols ⇒ Glycerol and fatty acids

2. In the **second stage**, these smaller molecules are converted into a limited number of central metabolic intermediates. Many sugars, fatty acids, and amino acids ultimately give rise to **acetyl-CoA**, whereas other amino acids enter metabolism as pyruvate or as intermediates of the citric acid cycle.

3. In the **third stage**, the acetyl group of acetyl-CoA is completely oxidized to CO_2 in the citric acid cycle. The electrons released during this oxidation are transferred mainly to NAD^+ and FAD, producing NADH and $FADH_2$. These reduced electron carriers subsequently donate electrons to the **respiratory chain**, and the energy released by electron transfer is used to generate ATP through oxidative phosphorylation.

In the first stage of catabolism, little or no energy is conserved. In the second stage, a relatively small amount of ATP is produced, and energy is also conserved in acetyl-CoA and reduced electron carriers. In the third stage, acetyl-CoA is oxidized in the citric acid cycle, generating NADH and $FADH_2$. These electron carriers donate electrons to the respiratory chain, where most ATP is produced through oxidative phosphorylation. Therefore, **most of the ATP that is generated in the degradation of nutrients is formed in the third stage of catabolism**.



The energy captured during catabolism is used to:

- Drive the biosynthesis of macromolecules and other cellular components
- Perform mechanical work
- Drive the active transport of molecules across membranes
- Maintain, replicate, and express genetic information

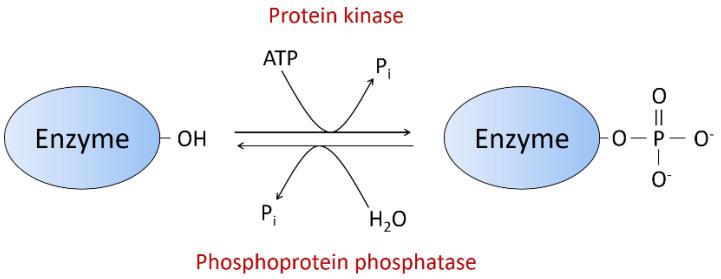
Regulatory mechanisms in metabolism

The flow of metabolites through a metabolic pathway is primarily determined by the activity of the enzymes involved. Therefore, metabolic pathways can be regulated by controlling the activity of their enzymes. In most pathways, regulation operates mainly through a small number of key enzymes.

The metabolism of a cell can be regulated in several ways:

1. Control of **enzyme abundance** through regulation of enzyme synthesis and degradation.
2. Control of the **catalytic activity** of the enzyme: there are several ways to regulate the activity of an enzyme.
    - Enzyme activity can be influenced by factors such as pH and the concentrations of substrates, products, and cofactors.

- One of the most important regulatory mechanisms is **allosteric regulation**. An allosteric enzyme changes its catalytic activity in response to stimulatory or inhibitory molecules. Allosteric regulators are often metabolites from the same pathway. For example, an end product may inhibit an enzyme acting early in the pathway, a mechanism known as feedback inhibition. The enzyme can also be regulated by metabolites from other pathways, allowing the activity of the pathway to be adjusted to the needs of the cell.
- Another type of enzyme activity control is **reversible covalent modification**. Covalent modifications often produce longer-lasting changes in enzyme activity than allosteric interactions. A major example of reversible covalent regulation is protein **phosphorylation**.
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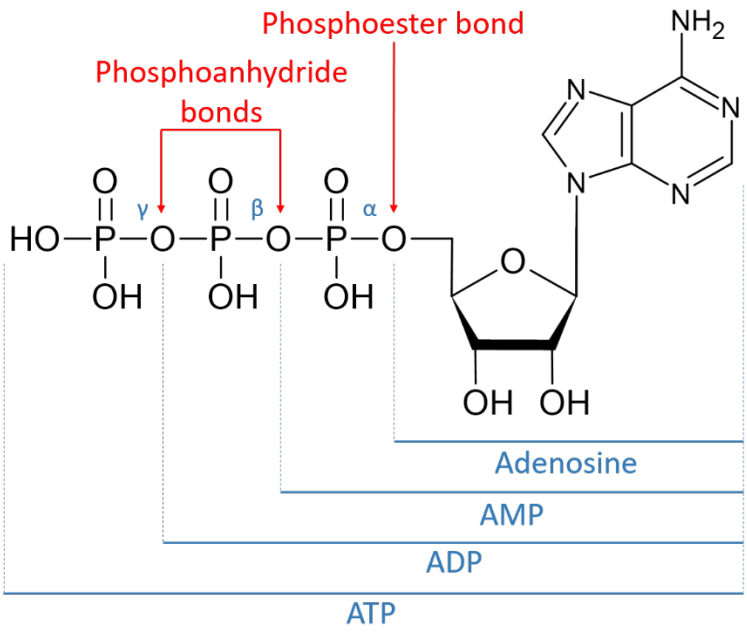
3. **Compartmentalization** of enzymes and substrates: different metabolic pathways occur in different cell compartments. Transporters control the movement of metabolites between compartments and can therefore influence their metabolic fate. For example, fatty acid synthesis occurs mainly in the cytosol, whereas fatty acid degradation by β-oxidation occurs mainly in the mitochondrial matrix. Transport of long-chain fatty acids into mitochondria is therefore an important regulatory step.

Together, these regulatory mechanisms continuously adjust metabolism to the cellular requirements for energy, metabolic intermediates, and end products. In multicellular organisms, extracellular signals such as hormones coordinate the metabolism of different tissues with the needs of the whole body. Metabolism is therefore continuously active and dynamically regulated.

Cells regulate the use and storage of nutrients according to their **energy requirements** and the **availability of metabolic resources**. In humans, energy is stored mainly as triacylglycerols in adipose tissue and as glycogen in the liver and skeletal muscle. The cellular **energy state**, reflected in the **relative concentrations of ATP, ADP, and AMP**, is a major factor in the regulation of metabolic pathways.

The role of ATP in metabolism

ATP, adenosine triphosphate, has a central role in metabolism. It is often described as the "energy currency" of the cell because ATP hydrolysis can be coupled to energetically unfavorable reactions and cellular processes.



The importance of ATP as an energy carrier arises from the large decrease in free energy associated with the hydrolysis of its phosphoanhydride bonds. This favorable free-energy change reflects the greater stability of the hydrolysis products rather than the release of energy simply through bond cleavage.

$$ATP + H_2O \rightarrow ADP + P_i \qquad \Delta G^{o\prime} = -30.5\ kJ/mol$$

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